Targeting the Elderly: from Spoofing to Persistence

Axelle Apvrille, Fortinet

Virus Bulletin, October 2026

Who am I?

Fortinet

- Principal security researcher with **Fortinet**
- Topics: **mobile malware** (Android, iOS) and **IoT malware**
- Founder of Ph0wn CTF, smart devices CTF (march 2027)

Since August 2026: **Reserve Squadron Commander** in the **French National Gendarmerie**.
This work was carried out *prior* to my appointment, and was conducted *entirely independently* of the Gendarmerie.

The investigation

Victim

All personal details of the ID are fake.

- **89** year old French man
- Victim of a social engineered **bank fraud**

Volunteer no.1

- On-site
- Linux ✓
- Case is on Windows... ✗
- Strong developer experience ✓
- Forensic ✗

Volunteer no.2

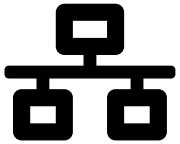
AI-generated picture. Surely I don't look like this, right?

- Remote
- Linux ✓
- Case is on Windows... ✗
- Security ✓
- A few forensic skills

The case is not technically advanced

no 0-day, no exploit, no rootkit

Defeated by social engineering



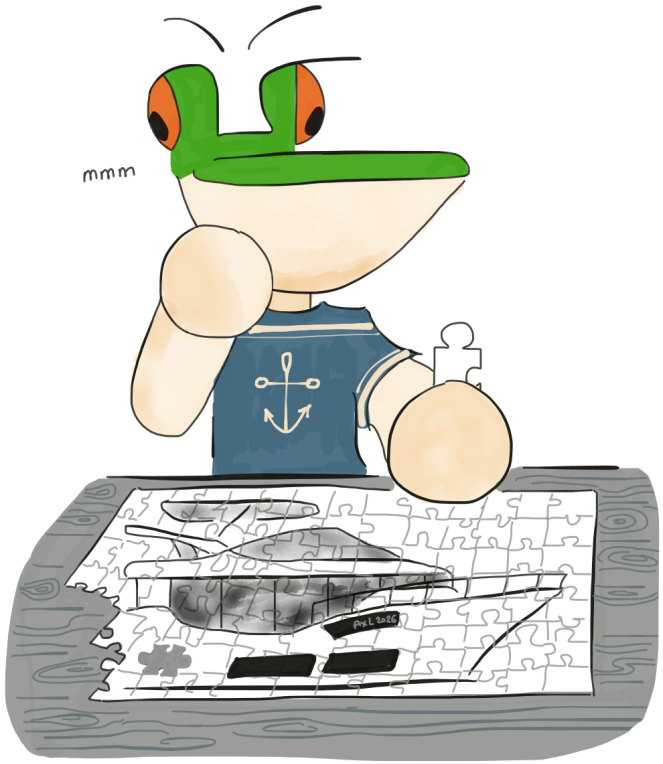
NAT

Password Manager

Anti-Virus

2FA

What's interesting in this case?



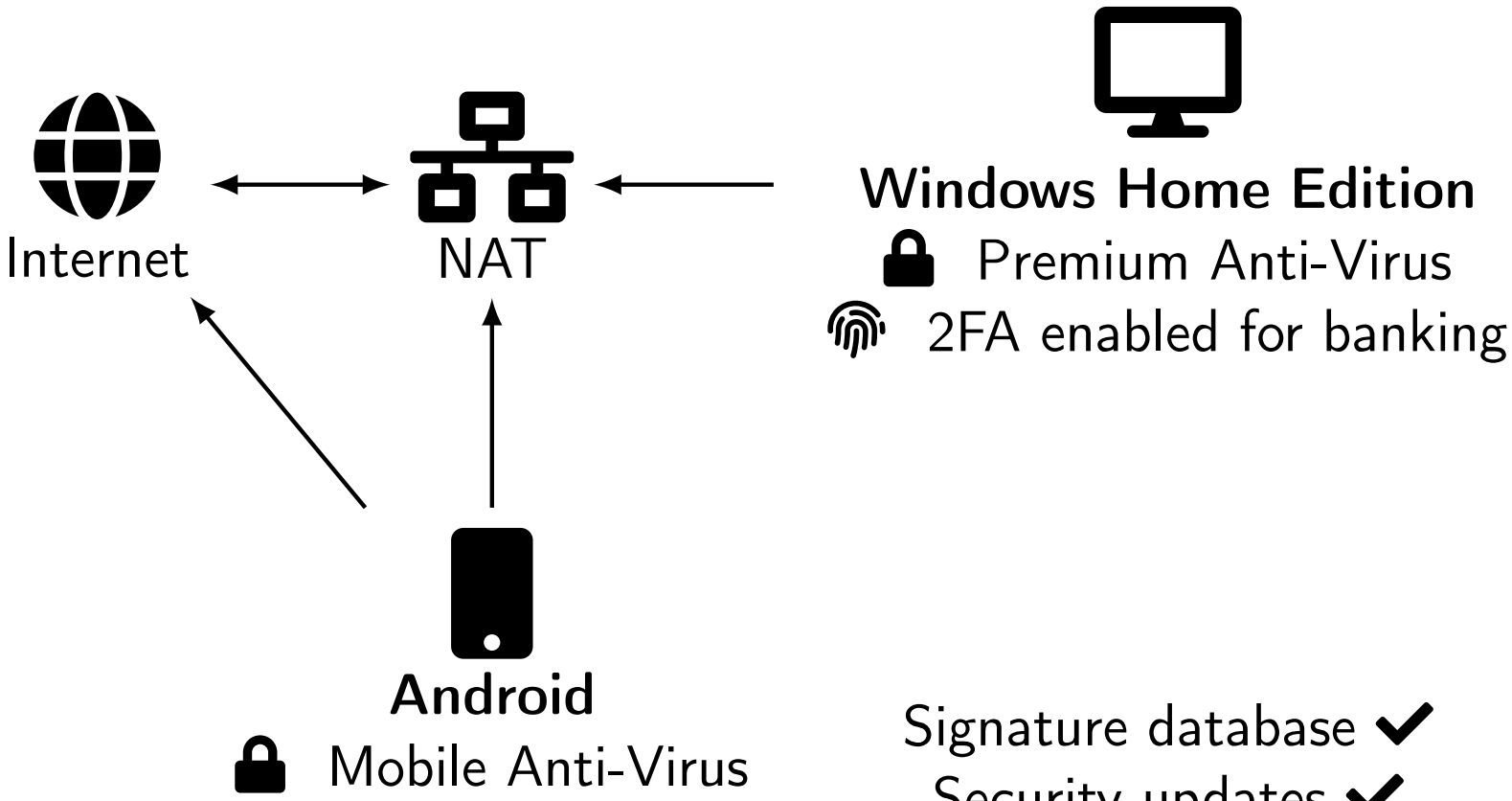
Pico le Croco is a personal hobby: hand drawn **crocodile** character, no AI, 100% fun.

1. **Full access**. Post mortem access to laptop, bank accounts, police report.
2. **Puzzle**. Re-constructing an attack from data, evidence... and uncertain evidence.

Age is an important factor

- He is **89 year old**.
- He is **not comfortable** with computers.
  SMS, WhatsApp, Email, Internet,
  Providers are **unclear notions** to him.
- His **memory is impaired**.
  No underlying medical condition has been identified.
- His **vision, hearing and walking** are consistent with his age.
- No guardianship, above average IQ.

IT environment



How it all started

--- Regarding the facts ---
--- On 03/02/2026, I received a notification on my mobile phone from my bank XXXX informing me that two direct debits had been made from my account on the same day.---
--- Later that day, an unknown number called me.--
--- A woman introduced herself as an advisor from my bank XXX Fraud Department.---

Translated from the French police report, which was filed on 05/02/2026

Step 1: Notification of fraudulent payment

- Notification? **We couldn't find any trace**.
- He can't remember where: WhatsApp, SMS, Email...
- Serious doubts this event occurred *before* the phone call from the attacker (impaired memory, uncertain chronology).

Step 2: Phone call from attacker

- February 3, 2026, 12:30 phone call
- Woman poses as **bank employee, from Fraud Department**
- She knows victim's full name + bank
- Her French is excellent
- "Unknown caller" : phone number probably not spoofed - but we can't be certain. Call not found in history: strange, was it on his landline?

Bank Fraud in UK in 2025: 65 million euros

54 Spanish phone fraudsters arrested in 2024

Operation PANDORA shuts down 12 phone fraud call centers

Sources: UKFinance Report, Europol 2024, Operation Pandora.

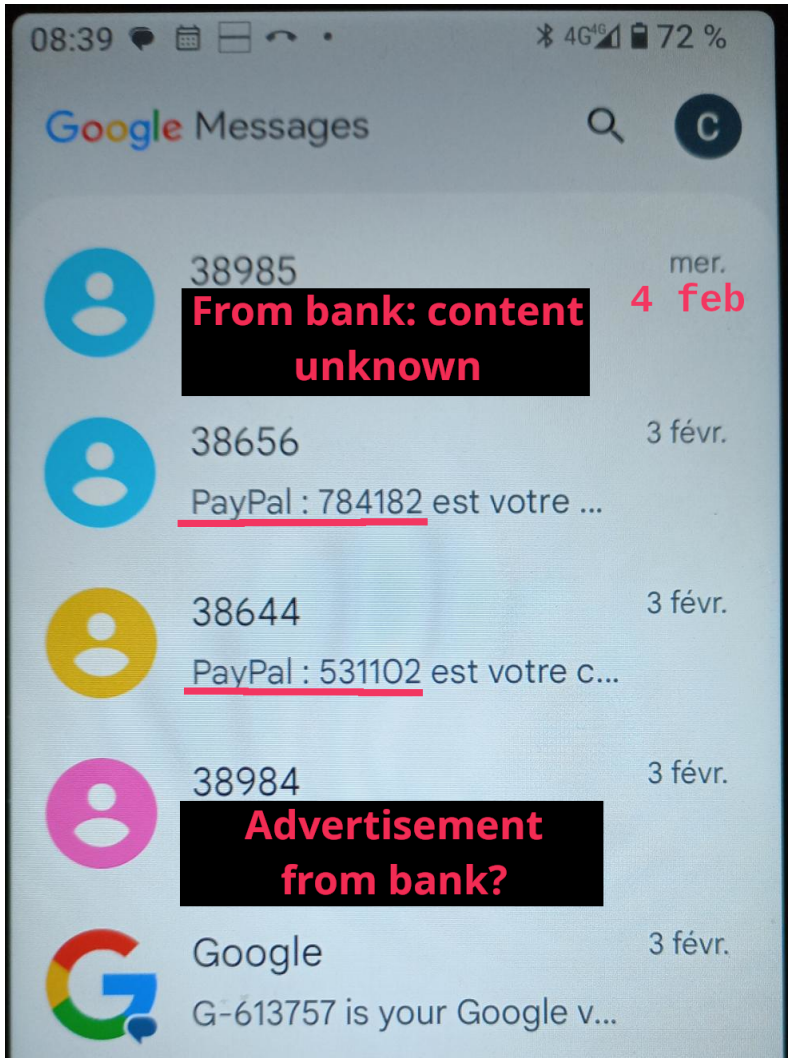
Step 3: Purchases

Counter a scam with another scam

CENSORED POLICE REPORT EXCERPT

- The amount is wrong. It was 1250 euros.
- She asked, so he gave: **typical for seniors**.
- From victim's initial point of view, she had **authority**, and she was **helping** him.

Bye, bye, PayPal 2FA security



- The woman said the purchases weren't real, and wouldn't be *debited*.
- The victim *trusted* the woman.
- He provided his **credit card** details, and **read the 2FA** to the woman...

Yes, we nearly receive alerts about such frauds every day...

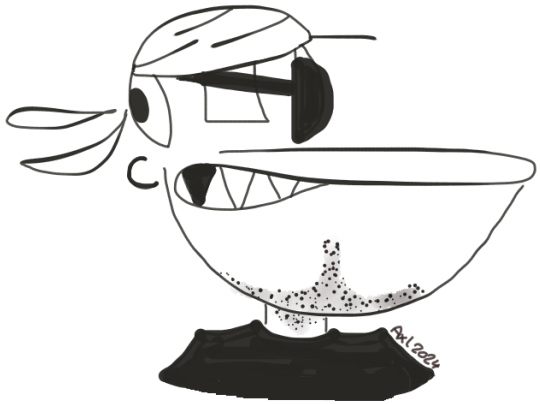
The last Google 2FA SMS is suspicious too, because the victim doesn't use GMail any longer.

PayPal transactions

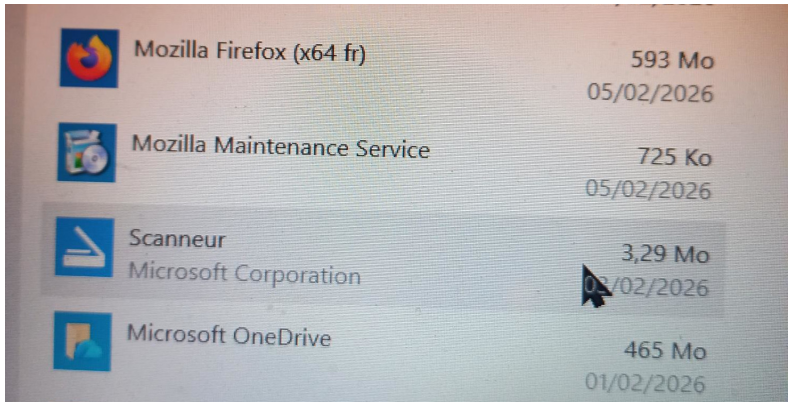
CENSURED IMAGE

- **Kiwi.com**: travel agency, headquarters in the *Czech Republic*.
- **Notino**: *Czech* perfume and cosmetics retailer.
- Attacker in the Czech Republic? Possible, but **this is not a proof**.

Step 4: Identity theft

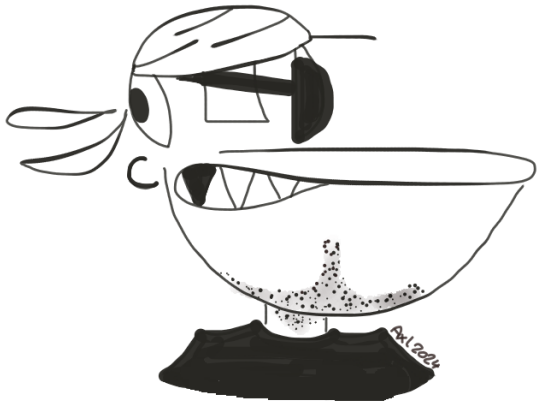


- Attacker registers a Western Union account under victim's identity.
- An ID card is necessary.
- Attacker instructs victim to install **Microsoft Scanner** and scan his identity card.



Scanning application installed on February 3, 2026

Step 4: Identity theft



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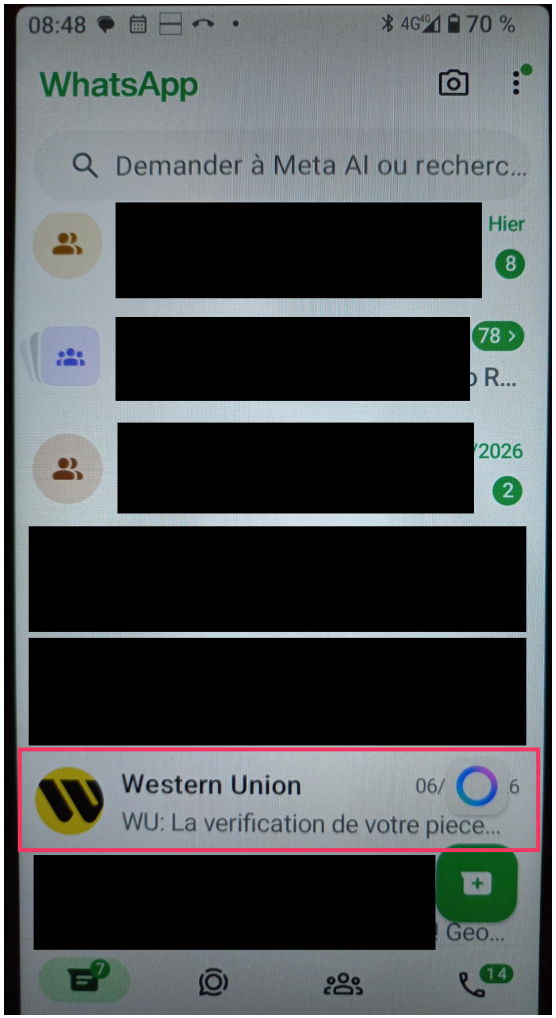
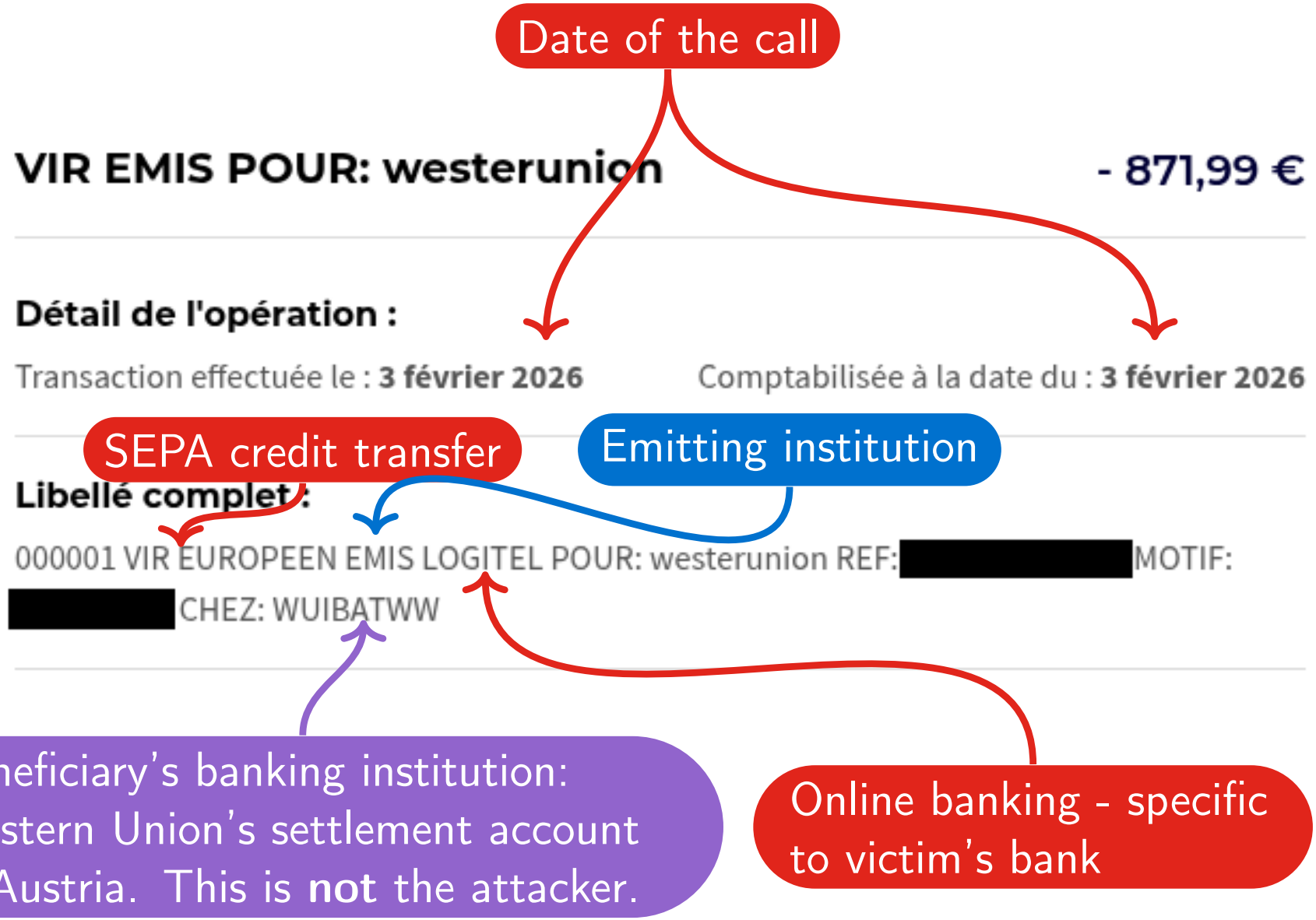


WU *Western Union*: Verification of your ID was succesful

SEPA credit transfer

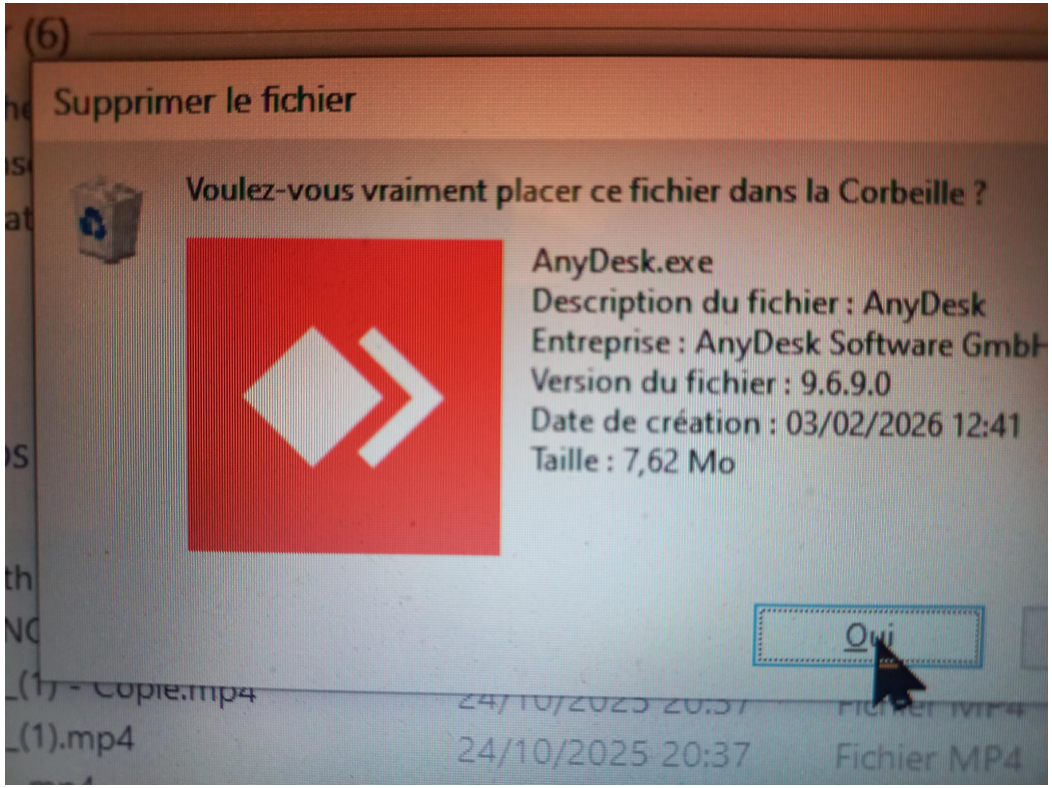


Attribution is impossible with our material

- PayPal purchases were done in shops related to the **Czech Republic**.
- Western Union transfers went to **Austria**.
- The airline ticket ✈ was for **Tunisia**.
- The woman spoke perfect **French** 💬.

Pay attention not to attribute the attack to a country.
We do not have the right evidence for that.
Attacker could be from *any country*

Step 5: Persistance



- Windows Home Edition does not support **RDP** for remote desktop access.
- **VNC** is not convenient because the victim is behind a NAT gateway.
- Attacker instructed victim to install **AnyDesk** at 12:41 🕛, and provide his AnyDesk ID.

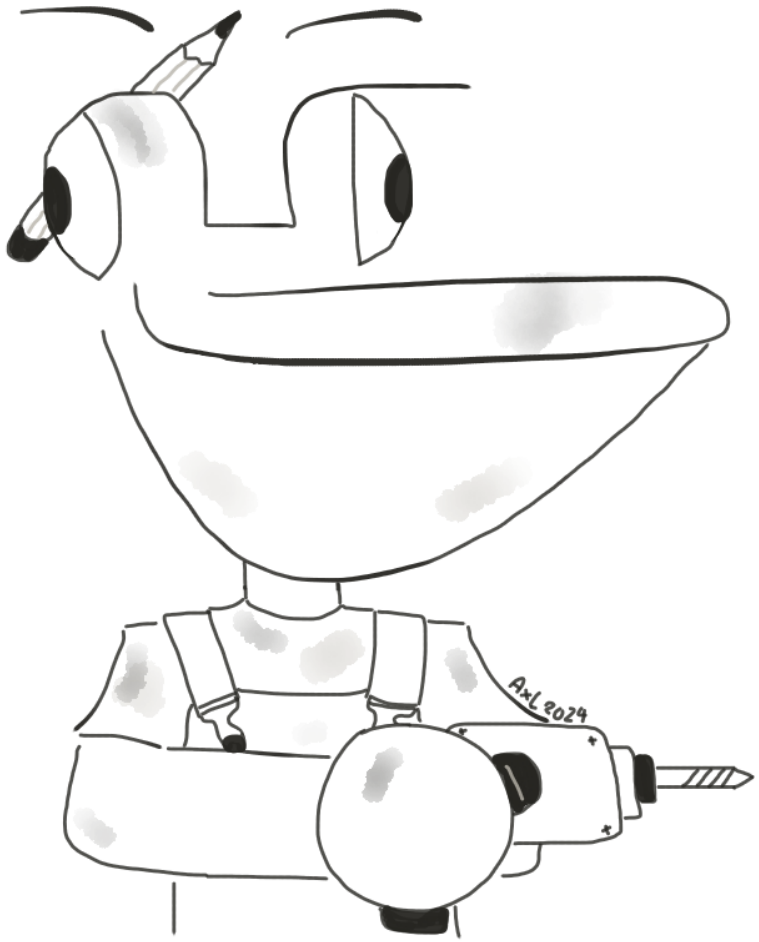
Attacker has full remote access to laptop.
Strangely, seeing his mouse move “by itself” alerts the victim, who “disconnects” his laptop.

Several other applications were installed

- Victim does not remember installing them (but his memory is bad).
- He had **no apparent reason** to install 2 different browsers.
- **Likely scenario: attacker installed** them during a remote AnyDesk session on February 5, 2026
- **AnyDesk session history was erased**. No trace of the scanned identity card either.
- After verification, **the installed apps were genuine, not trojaned**.

Remediation steps

- A **full OS re-install** + **password changes** would have been recommended!
- But it's **difficult at 89**.
- Uninstalled AnyDesk and all recent applications. Re-installed anti-virus from trusted source.
- Creation of **standard Windows account** for the victim, and **admin accounts** for his IT help.
- Credit card blocked, ID card re-issued, meetings with bank.



Summary of Losses

3500 euros

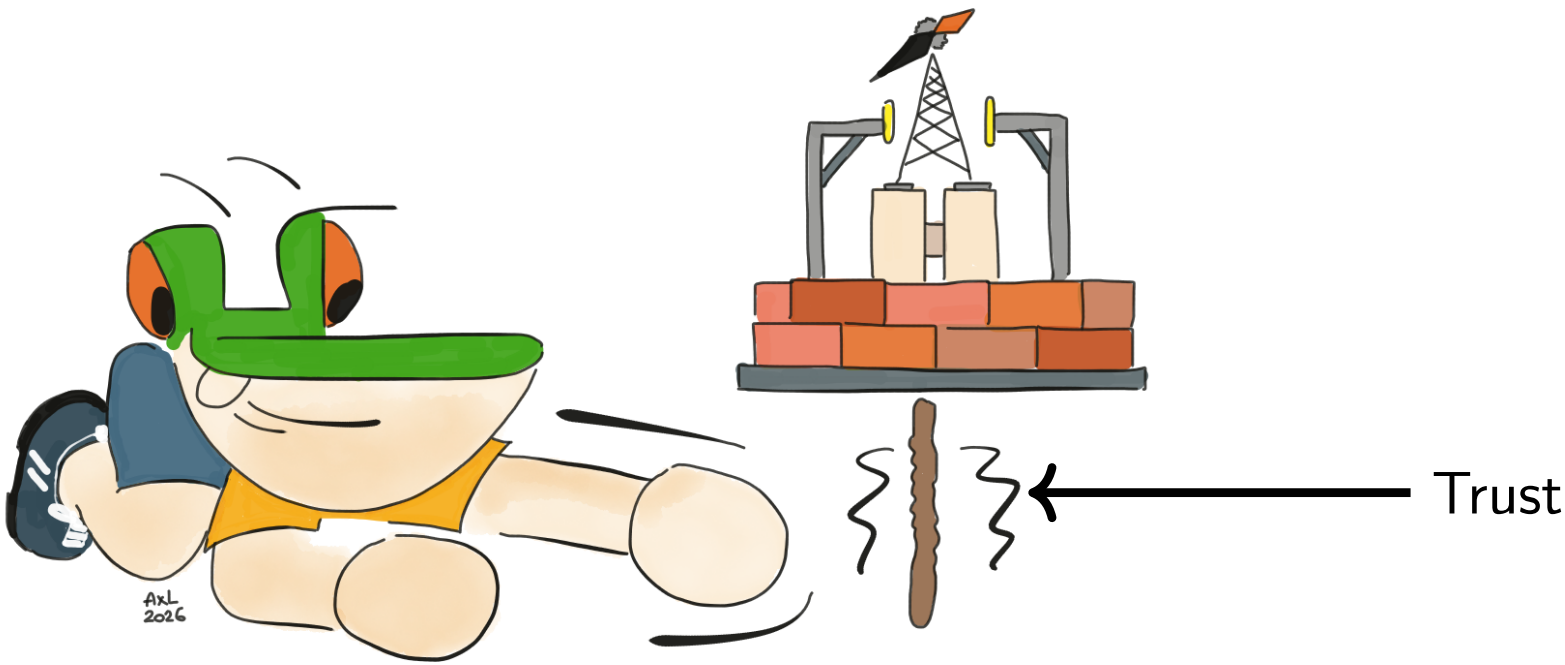
1 Western Union payment did not go through: 900 euros ✔
No more banking hemorrhage ✔
Personal files may have been collected ✘

How frequent is this situation?

Age range	Population percentage in France
Over 80	6%
10-16	9%
Total	**15%**

We have built systems which fail to protect 15% of the population!

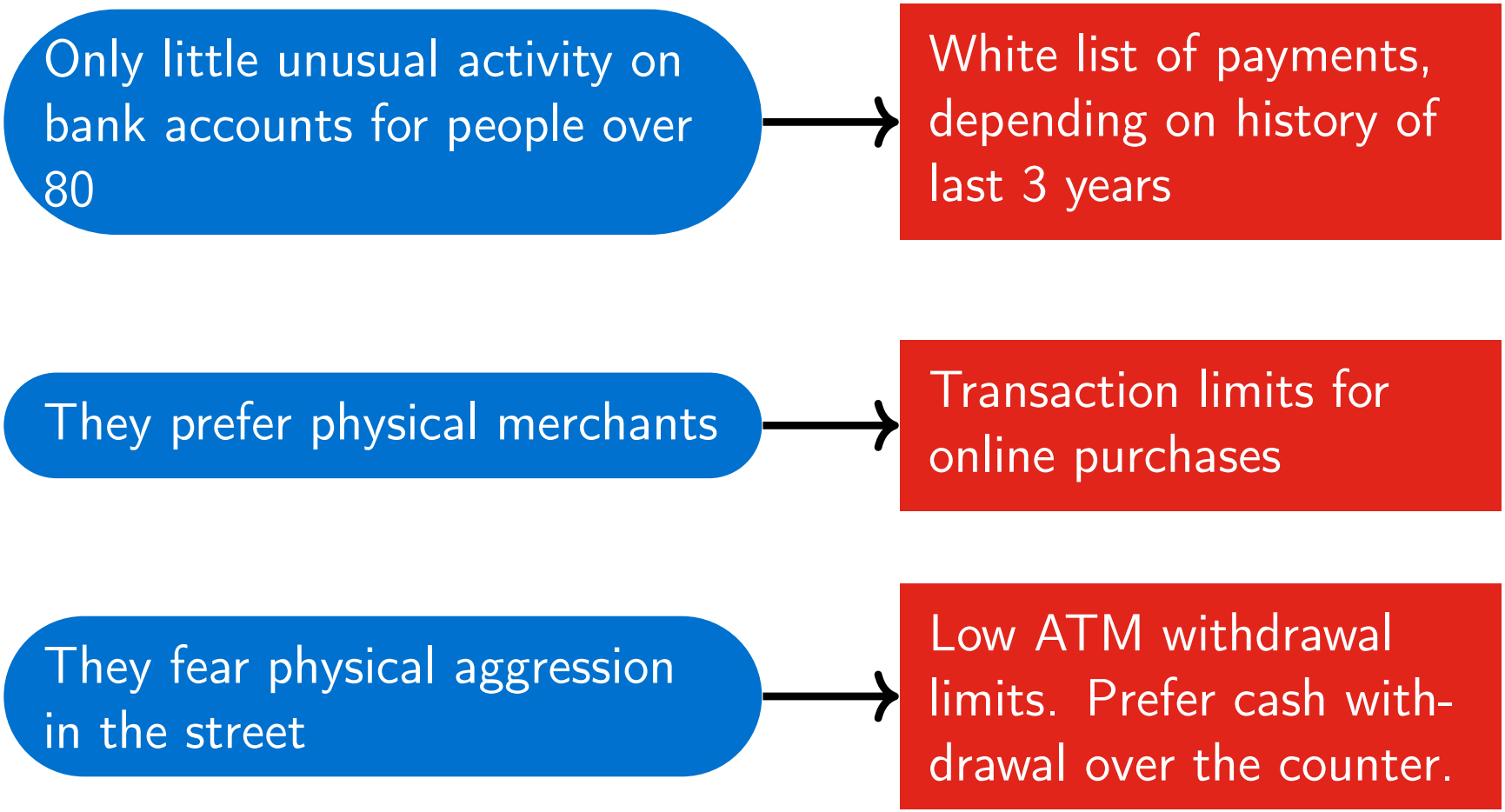
What security relies on



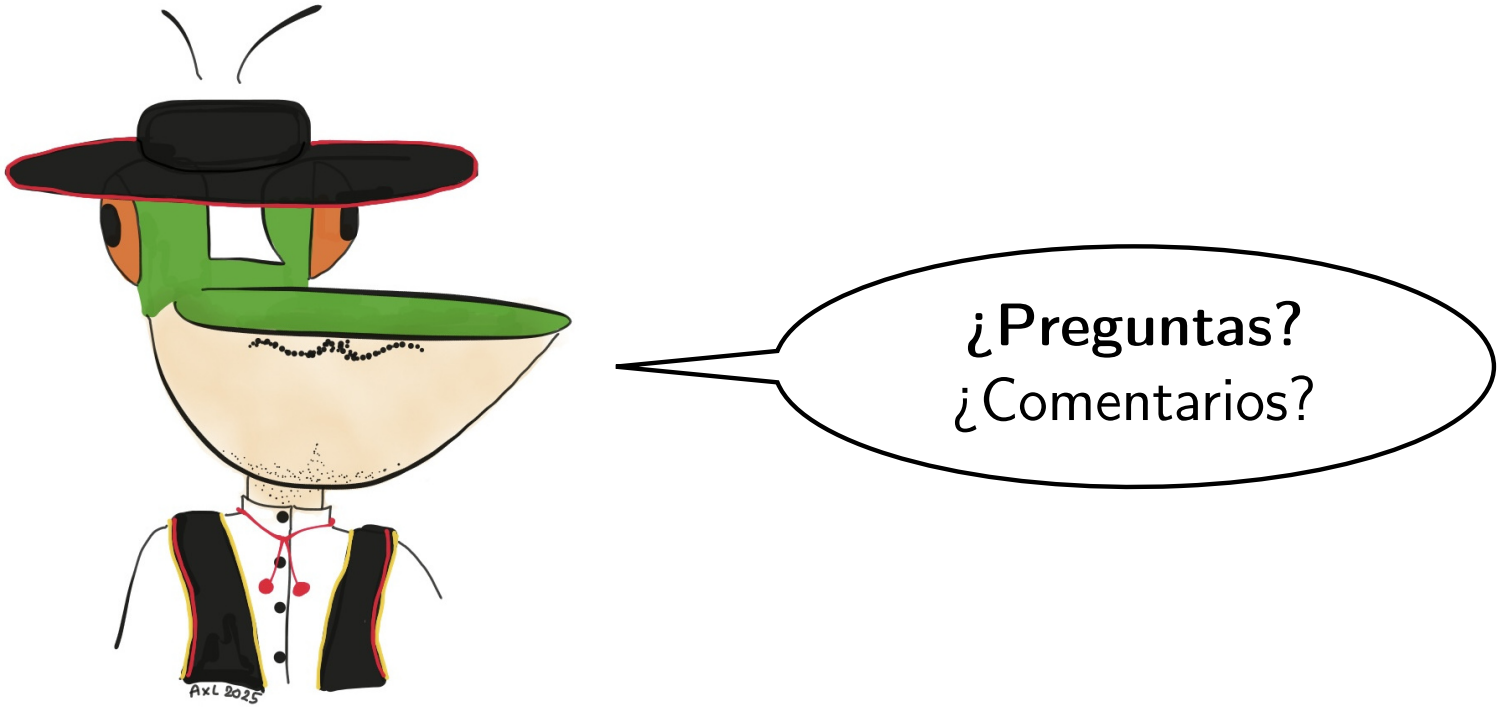
Operational bank fraud detection systems

- SAS Financial Crimes Analytics, TietoEVRY BankTech Fraud Prevention, EBA Clearing Fraud Pattern and Anomaly Detection, Société Générale ML System, MasterCard Decision Intelligence...
- Behavioural profiling: transaction history, device and connection patterns, location...
- Collaborative threat intelligence.

Solutions


Only little unusual activity on bank accounts for people over 80
White list of payments, depending on history of last 3 years
They prefer physical merchants
Transaction limits for online purchases
They fear physical aggression in the street
Low ATM withdrawal limits. Prefer cash withdrawal over the counter.

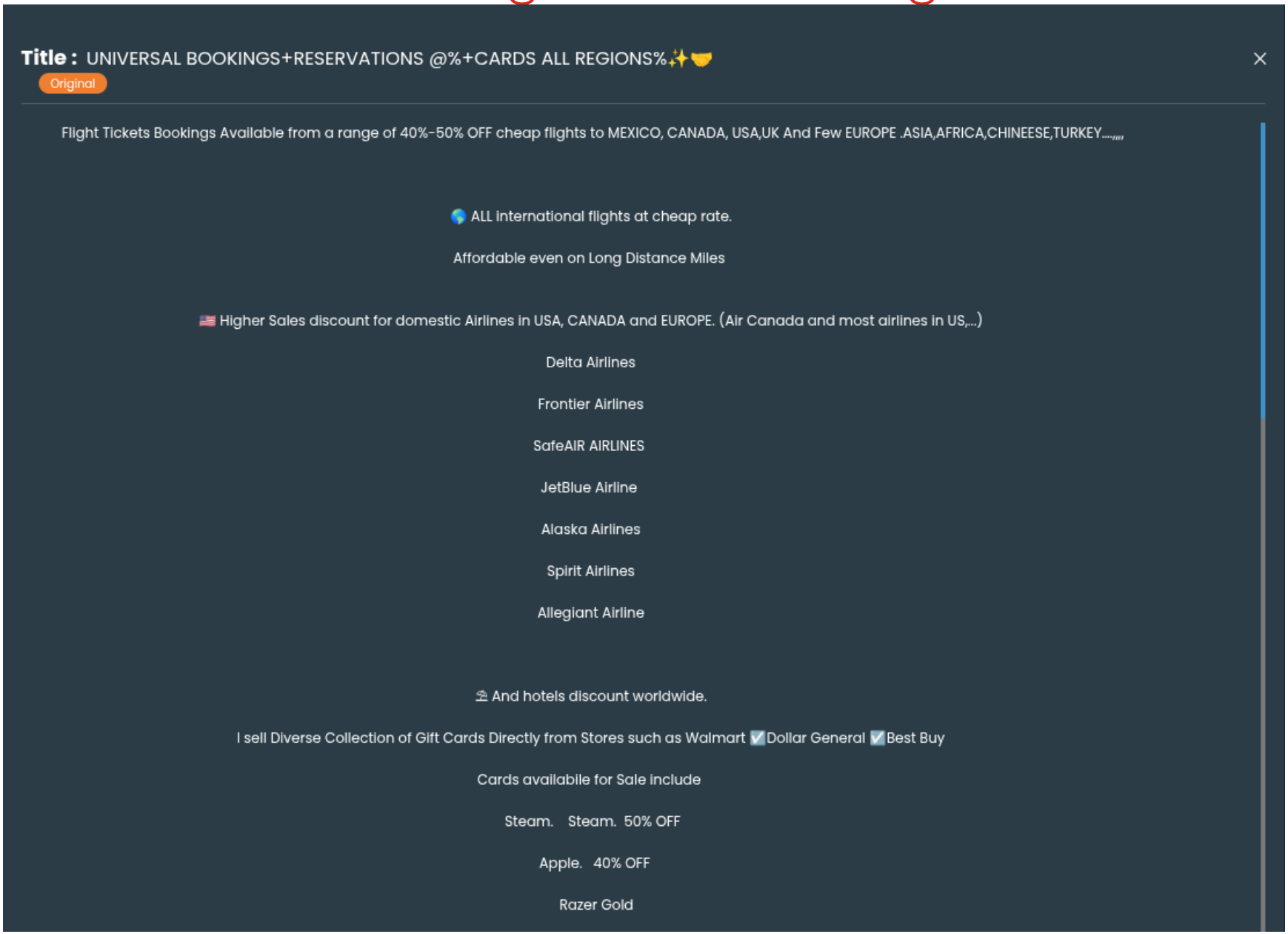
Thanks for your attention. Questions? Share ideas?



- Cryptax on Mastodon, BlueSky, Discord
- `aapvrille@fortinet.com`
- https://fortinet.com/blog/threat-research
- FortiGuard Publications
- Photo credits: magnific, Tom Morbey, Christina at wocintechchat.com, Sei, Shifaaz Shamoon on Unsplash

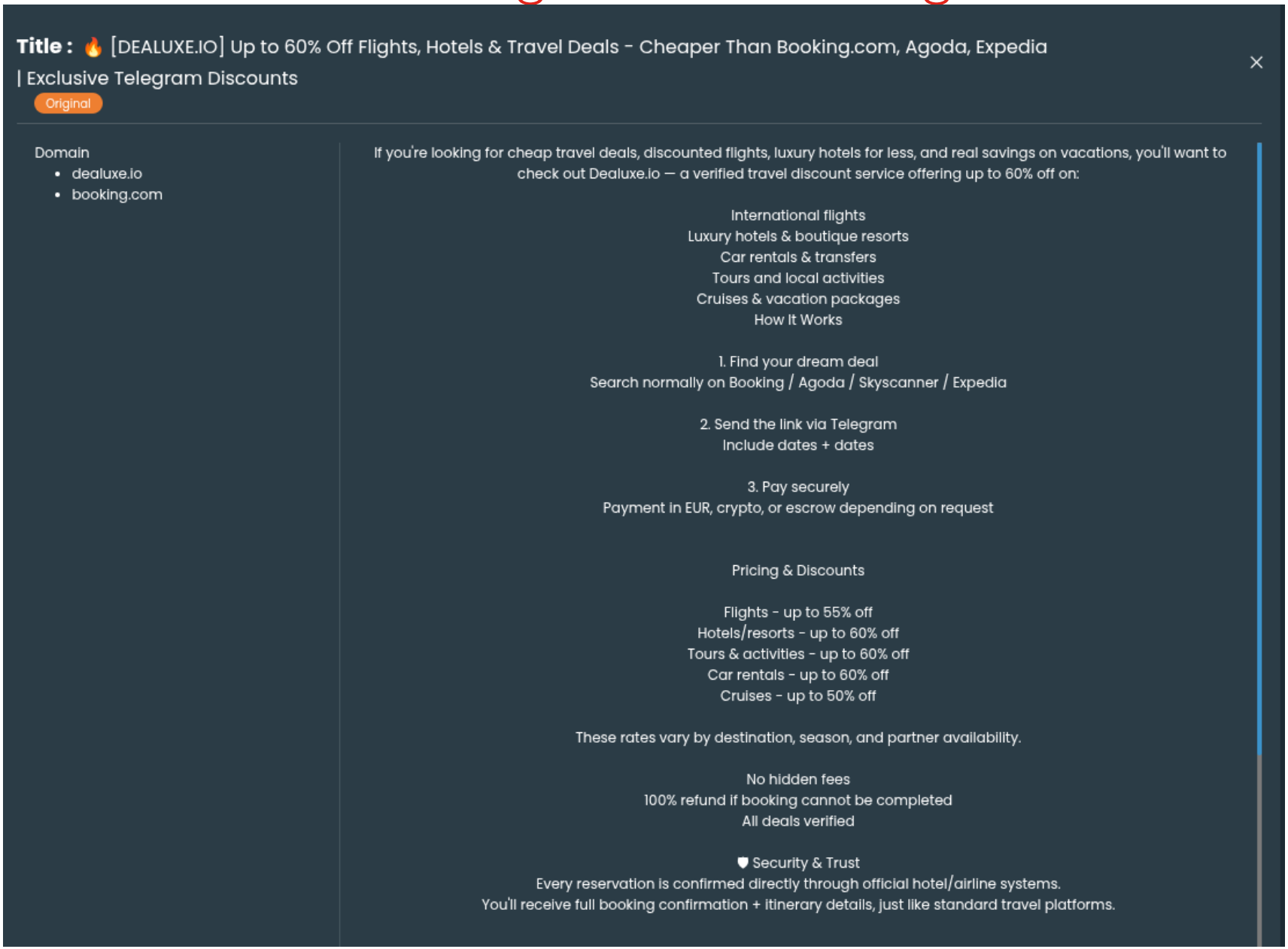
Backup slides

Dark Forums Discount flight ticket bookings



Underground forums, Sept - Nov 2025

Dark Forums Discount flight ticket bookings



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Original

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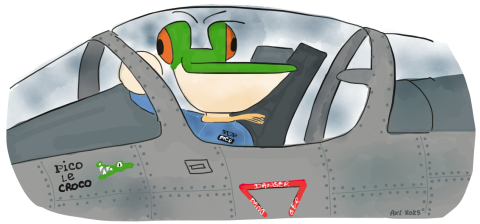
Underground forums, Sept - Nov 2025

Airline ticket fraud



Frequent questions

- The victim purchased an airline ticket. How does the attacker **convert it to money**?
- Is the name on the ticket the name of the attacker? **Generally, no**

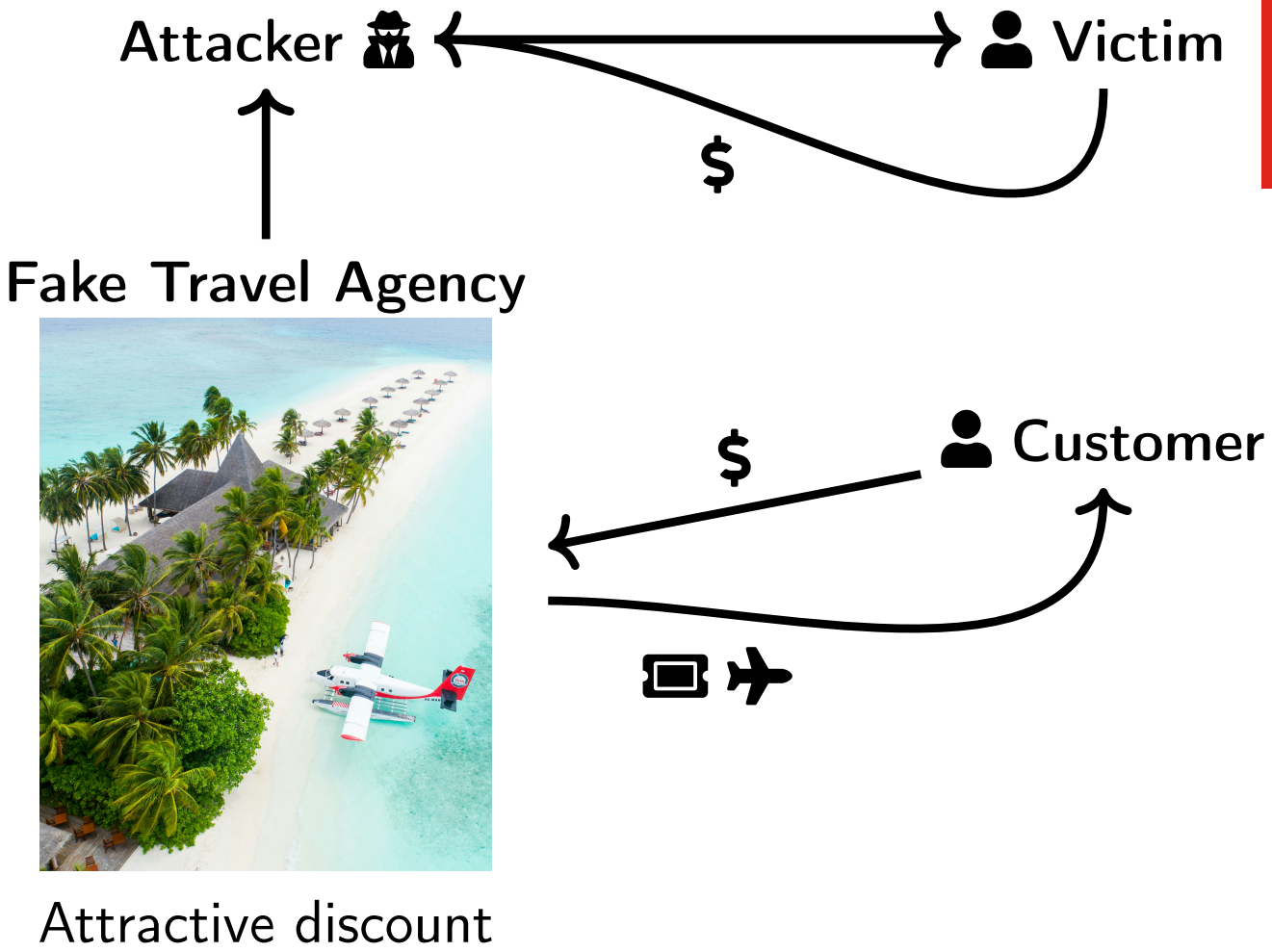
References:

- Hutchings, A., Crime Law Soc Change 70, 461-487 (2018)
- Interpol, Airline ticket fraud
- The Wall Street Journal, Dark Web Travel Agencies Take Flight, 2025

Fake travel agencies

Keeps customer money

Attacker poses as bank employee. Lures victim to pay flight ticket



Attractive discount

Customer purchases flight ticket at attractive price